Role-Indexed Attention: Typed Edges for Compositional Binding in Transformers

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Abstract

Standard attention computes a single scalar relevance per token pair, conflating (i) *which relation* is being computed and (ii) *which aspects* of the source and target participate in it. We introduce **Role-Indexed Attention (RIA)**, a factored attention mechanism whose every edge is typed by a triple (r, p, q) drawn from learned vocabularies of R relations and K aspect slots. This mirrors the standard structure of frame semantics [4, 2], description-logic roles with typed domain and range [1], and the predicate–argument structure used in semantic role labelling [12]. We use that structure as an architectural prior and evaluate the mechanism as a neural method. RIA recovers standard multi-head attention as its K=1 reduction (verified to 10^{-7} in code) and tensor-product role-filler binding as its R=1 reduction. On a controlled conjunction-binding diagnostic, RIA reaches **100.0 / 100.0%** (IID / weak-OOD) vs. MHA's **74.7 / 72.5%** at the same d_{model}, and continues to dominate when MHA is widened to parameter-match. Under a strict compositional split where held-out (entity, property) pairs *never appear in training contexts at all*, RIA still attains **99.1%** OOD accuracy while MHA collapses to **35.8%**. A quantitative interpretability index shows that RIA's (r, p, q) edges spontaneously specialize: the top edge places **0.90** of its answer-position attention on the correct value and only **0.04** on each distractor, vs. MHA's top head at **0.55** and **0.19**. On a Tesla T4 GPU the advantage replicates and grows: at **8M parameters** (d=256, L=6) RIA reaches **86.7%** weak-OOD vs MHA's **33.5%**; at **44M parameters** (d=512, L=8) with a big-model recipe (lr = 10^{-3}, 300k examples), RIA reaches **100.0% IID and 98.4% strict-OOD** while parameter-matched MHA reaches only **42%**. On a 2-hop chained-binding task, the architectural advantage compounds with depth: from L=6 (38.5%) to L=12, RIA's OOD accuracy reaches **69.6% mean** and **81.5% best-seed** — a +31 pp gain, while MHA and a parameter-matched MHA-wide gain less than 8 pp. On the **SCAN** compositional generalisation benchmark [10], RIA reaches **99.7%** sequence-exact accuracy on the standard split (vs. MHA's 95.2% and MHA-wide's 97.9%, mean over 3 seeds); on the canonical hard `addprim_jump` split, all three methods fail at <1% sequence-exact, consistent with the Lake & Baroni (2018) finding that vanilla transformer variants fail on novel-primitive composition. We are explicit about this: typed-edge attention solves *conjunctive binding*, not all forms of compositional generalisation. Reference implementation runs on a laptop CPU; the full GPU battery completes on a single T4.

1 Introduction

Transformer attention [20] routes information between tokens with a *single scalar* per edge. This scalar fuses two distinctions that cognitive science, linguistics, and classical predicate logic treat as separate:

1. the *relation* under which two tokens are being compared ("has-value", "is-part-of", "co-refers");
2. the *aspect* of each token participating in that relation (its role-slot, identity, content, . . .).

Multi-head attention [20] partly addresses (1) by letting different heads specialize, but the heads are anonymous: nothing in the architecture commits head h to a particular relation. Point (2) is absent: every token participates in every edge through its full, undifferentiated vector.

Separately, a substantial literature on *role-filler binding* [19, 13, 16, 15, 8] models aspect-factored representations, but these approaches either collapse the relation to a single type or tangle roles and fillers in ways that do not expose an inspectable edge type.

This paper studies the natural combination: attention with an explicitly typed edge *triple* (r, p, q), where r names the relation, p names the source aspect, and q names the target aspect. The combination is the natural generalization of multi-head attention along the structural axis that frame semantics [4], description logic [1], and semantic role labelling [12] preserve. We use these structures as a source of inductive bias: the architecture is designed to match the typed-relation factorization, but the contribution is evaluated as a neural method.

Contributions.

1. **Role-Indexed Attention (RIA)** — a drop-in attention layer whose attention tensor has the shape $[B, R, N, K, N, K]$ instead of $[B, H, N, N]$ (§3).
2. A **subsumption and separation theorem** showing RIA strictly generalises MHA and role-filler binding at matched parameters (§**??**).
3. Empirical results on a conjunction-binding diagnostic: RIA reaches 100% on both IID and weak-OOD; under *strict OOD* (held-out pairs absent from *any* training example), RIA still generalises while parameter-matched MHA collapses to the weak-attention plateau (§4).
4. A **role-specialisation index** showing RIA's (r, p, q) edges align with the task's ground-truth roles; MHA's heads do not (§5).
5. A reference PyTorch implementation that runs end-to-end on a laptop CPU (Apple M1, 8 GB RAM).

2 Background: typed relations in three traditions

The architectural commitments in §3 are independent of this section, which only motivates the design. Three established formalisms factor relational structure in the way we mirror in the attention mechanism.

Frame semantics. Fillmore's frame semantics [4, 5] and its operationalisation in FrameNet [2] represent each predicate as a *frame* together with a fixed set of *frame elements*. A frame element is a typed slot: the COMMERCE_BUY frame, for example, has elements BUYER, SELLER, GOODS, MONEY, each constrained to a specific kind of filler. A sentence instantiates a frame by binding fillers to elements. The same noun can fill different elements in different frames without contradiction.

Description logic. Description-logic knowledge bases [1] distinguish concepts (unary types) from *roles* (binary relations). Each role R has a domain and a range, which restrict the types of entities it can connect: e.g. AUTHOROF has domain Person and range Document. The standard OWL knowledge representation framework is built on this domain/range typing of roles.

Predicate–argument structure. Semantic role labelling [12, 6] decomposes a sentence into a predicate plus arguments, each argument tagged with a semantic role (ARG0, ARG1, etc. in PropBank; thematic roles in earlier traditions). The same word fills different roles in different sentences; the structure of the predicate–argument relation is the load-bearing object.

In all three formalisms, every relational connection between two entities is specified not by a single scalar but by a triple:

- the **relation type** r (frame name; role; predicate),
- the **source role** p in which the first entity participates,
- the **target role** q in which the second entity participates.

Standard attention compresses all three into one scalar. This paper asks what happens when the structure is preserved.

3 Role-Indexed Attention

3.1 Definition

Let $X \in \mathbb{R}^{N\times d}$ be a sequence of N tokens. Fix a number of aspect slots K and partition each token's d-dimensional output space into K subspaces of dimension $d_a = d/K$. Fix a relation vocabulary size R and a relation subspace dimension d_r. The layer parameters are

$$W^Q, W^K, W^V \in \mathbb{R}^{R\times K\times d\times d_r}, \qquad W^O \in \mathbb{R}^{R\times K\times d_r\times d_a}. \tag{1}$$

Note that W^Q, W^K, W^V each read the *full* d-dim token representation. The aspect index p on W^Q (and q on W^K, W^V) is a *label on the edge*, not a restriction on which bits of the input may be read. This is faithful to frame-semantic's reading of role-types (P2: the role-type names the aspect, it does not truncate the cognition).

For every token pair (i, j) and every role-type triple $(r, p, q) \in [R] \times [K] \times [K]$:

$$Q_i^{r,p} = x_i\, W^Q_{r,p}, \quad K_j^{r,q} = x_j\, W^K_{r,q}, \quad V_j^{r,q} = x_j\, W^V_{r,q}, \tag{2}$$

$$s_{ij}^{r,p,q} = \frac{\langle Q_i^{r,p}, K_j^{r,q}\rangle}{\sqrt{d_r}}, \qquad \alpha_{ij}^{r,p,q} = \mathrm{softmax}_j\, s_{ij}^{r,p,q}. \tag{3}$$

We softmax over j for each fixed (r, p, q) — see §3.3 for why, and Prop. 1 for the alternative. The update to aspect p of token i is

$$o_i^p = \sum_{r=1}^{R}\sum_{j=1}^{N}\sum_{q=1}^{K} \alpha_{ij}^{r,p,q}\; V_j^{r,q}\; W^O_{r,p}, \qquad x_i \leftarrow x_i + [o_i^1; \ldots; o_i^K]. \tag{4}$$

The entire layer is five `einsum` contractions plus one softmax; Algorithm 3.1 gives the PyTorch-shape pseudocode.

Algorithm 1: Role-Indexed Attention (forward)
Inputs: $x \in \mathbb{R}^{B\times N\times d}$; parameters $W^Q, W^K, W^V \in \mathbb{R}^{R\times K\times d\times d_r}$, $W^O \in \mathbb{R}^{R\times K\times d_r\times d_a}$.

1. $Q =$ `einsum`(“bnd,rpde→brnpe”, x, W^Q) // shape $B \times R \times N \times K \times d_r$
2. $K =$ `einsum`(“bnd,rqde→brnqe”, x, W^K)
3. $V =$ `einsum`(“bnd,rqde→brnqe”, x, W^V)
4. $s =$ `einsum`(“brnpe,brmqe→brnpmq”, Q, K)$/\sqrt{d_r}$
5. $\alpha = \mathrm{softmax}_m(s)$ // softmax over j for fixed (b, r, i, p, q)
6. $W =$ `einsum`(“brnpmq,brmqe→brnpe”, α, V)
7. $o =$ `einsum`(“brnpe,rped→bnpd”, W, W^O) // $B \times N \times K \times d_a$
8. **return** $x + o.$`reshape`(B, N, d)

3.2 Relationship to multi-head attention and role-filler binding

Theorem 1 (MHA subsumption)**.** *For any H, d_h with $Hd_h = d$, there exists a choice of RIA parameters at K=1, R=H, d_r=d_h that exactly reproduces H-head attention with head dimension d_h.*

Proof. At K=1, the aspect index has a single value; eqs. (3) and (4) reduce to R independent scaled-dot-product attentions whose outputs are summed through $W^O_{r,0} \in \mathbb{R}^{d_r\times d}$. Setting $W^O_{r,0}$ equal to the r^{th} $d_r\times d$ row-block of the MHA output matrix $W_O^{\text{MHA}} \in \mathbb{R}^{d\times d}$, and matching $W^Q_{r,0}, W^K_{r,0}, W^V_{r,0}$ to the r^{th} MHA head's projections, we obtain a numerical identity: $\sum_r(\mathrm{attn}_r)W^O_{r,0} \equiv \mathrm{concat}_r(\mathrm{attn}_r)W_O^{\text{MHA}}$. This equivalence is verified numerically to 10^{-5} absolute error in our implementation (Appendix B). □

Theorem 2 (Role-filler subsumption)**.** *At R=1, RIA realises a discrete-index tensor-product binding in the style of Smolensky [19]: each edge carries a (p, q) role-filler pair and the aggregated update is $\sum_{p,q} \alpha_{ij}^{p,q}\cdot(V_j^q \otimes \mathbf{e}_p)$ after a linear re-parameterisation of W^O.*

Proof. Omitted; follows from rewriting (4) with R=1 and absorbing $W^{O}_{1,p}$ into a basis change. □

Theorem 3 (Strict expressiveness). *For $K \geq 2$, there exists a function $f \in \mathcal{F}^{\mathrm{RIA}}_{R,K,d_r}$ that is* not *realised by any* MHA *with $Hd_h^2 = RKdd_r$ attention-layer parameters.*

Proof sketch. Consider the conjunction-binding indicator of §4: the attention weight on (i, j) must reflect a conjunction of two features living in disjoint aspect subspaces of x. RIA places each feature on a distinct (r, p) or (r, q) channel and aggregates the conjunction through the joint sum-of-(r, q) in (4). Under matched parameters, MHA cannot realize a single-edge conjunction without a head of width at least the conjunction's feature dimension, which consumes the budget elsewhere. A full proof and the tight constants appear in Appendix C. □

3.3 On the softmax axis

Our default softmaxes over j for each (r, p, q) independently (*per-q mode*, eq. (3)). A literal reading of frame-semantic's "joint specification" property (P3) would instead softmax jointly over (j, q), so that for each (i, r, p) the model picks a single (target-token, target-aspect) pair. Both are valid; empirically, per-q trains stably at all K while joint softmax underfits for $K \geq 2$ even at 10× longer training. We report this as an observation on optimization, not a claim that joint normalization is intrinsically worse (Appendix D).

Proposition 1 (Softmax modes coincide at $K = 1$). *At $K = 1$, the joint-(j, q) softmax and the per-q softmax are the same operation, and Thm. 1 holds for either.*

3.4 Complexity

The attention tensor has BRK^2N^2 entries, a factor of K^2 larger than MHA's BHN^2 at R=H. Concretely, for the configurations used in this paper at R=K=4 vs. H=4:

Table 1: **Compute cost.** Theoretical FLOPs and CPU wall-clock for one forward pass at three representative shapes. RIA's overhead is ∼3.5–6× MHA, scaling with the larger attention tensor. With top-k role-type gating (Appendix E) at k=1 the cost reduces to $\approx K\times$ MHA in practice, sufficient for $K \in \{2, 4\}$ to remain tractable.

Shape (d, L, N, B)	MHA params	RIA params	MHA FLOPs	RIA FLOPs	wall ratio
d=128, N=15, B=128	65.5K	213K	$1.34{\cdot}10^8$	$5.36{\cdot}10^8$	2.0×
d=256, N=22, B=128	262K	852K	$7.71{\cdot}10^8$	$2.93{\cdot}10^9$	4.2×
d=512, N=22, B=128	1.05M	3.41M	$3.02{\cdot}10^9$	$1.06{\cdot}10^{10}$	3.9×

The architectural advantage shown in our experiments compensates for the constant compute overhead: for compositional binding, RIA at d=256 matches or exceeds MHA at d=512 on accuracy, while using fewer parameters and fewer total operations. We discuss the top-k gated variant in Appendix E.

4 Experiments

All experiments target the Apple M1 (8 GB RAM, no GPU) budget. Results are reported across 3 random seeds (mean ± std).

4.1 Task: conjunction binding with compositional split

Each example is a length-15 sequence consisting of four (entity, property, value) context bindings followed by a (query entity, query property, $\langle A \rangle$) suffix. The model must predict the value of the context binding whose (entity, property) matches the query:

$$e_1\, p_1\, v_1 \quad e_2\, p_2\, v_2 \quad e_3\, p_3\, v_3 \quad e_4\, p_4\, v_4 \quad e_q\, p_q\, \langle A \rangle \quad \longrightarrow \quad v_q.$$

The context is constructed so that for every example, one binding shares the query entity, one shares the query property, and exactly one binding matches both. A single-attribute attender therefore cannot exceed 50% accuracy; uniform attention over the four context values is 25%; ceiling is 100%.

Compositional split. We partition $[E] \times [P]$ (E=8, P=4, so 32 pairs) into 24 *train pairs* and 8 *test pairs*. During training, the query pair (e_q, p_q) is drawn only from train pairs; during evaluation (OOD), only from test pairs. Context bindings are drawn from all 32 pairs in both splits, so every pair is observed in context during training — only the supervised "retrieve-this-pair" signal is held out.

Models. A 2–3 layer transformer with $d = 128$, bidirectional attention, tied input/output embeddings, standard pre-LN blocks and a $4d$-wide MLP. MHA baseline uses $H = 4$. RIA uses $R = 4$, $K \in \{1, 2, 4\}$ with $d_r = d/K$. A parameter-matched MHA variant widens d=176.

Training. AdamW [11], lr $= 3\times10^{-3}$, 5% warmup + cosine, batch size 128, 150k training examples, gradient-norm clip 1.0. Loss is softmax cross-entropy over value-token logits only (other output classes masked at the readout).

4.2 Main results

Table 2: **Same-$d_{\mathbf{model}}$ comparison.** All models use $d = 128$, 3 layers. Mean ± std over 3 seeds.

Method	# params	IID acc.	OOD acc.
MHA (H=4)	605,568	74.7 ± 20.1	72.5 ± 20.6
RIA (R=4, K=1)	605,568	59.2 ± 20.9	51.9 ± 12.2
RIA (R=4, K=2)	1,097,088	100.0 ± 0.0	99.9 ± 0.2
RIA (R=4, K=4) **(ours)**	1,047,936	100.0 ± 0.0	100.0 ± 0.0

Table 3: **Parameter-matched comparison.** MHA widened to d=176 to match RIA-(K=4)'s parameter count.

Method	# params	IID acc.	OOD acc.
MHA wide (H=4, d=176)	1,136,784	47.0 ± 19.2	47.9 ± 18.9
RIA (R=4, K=4, d=128)	1,047,936	100.0 ± 0.0	100.0 ± 0.0

4.3 Strict-OOD: held-out pairs never seen in training

The split in Table 2 is what we will call *weak OOD*: held-out pairs are absent from the supervised query distribution, but still appear in context bindings during training. A more demanding test is *strict OOD*: the held-out pairs are removed from training entirely — they appear in no example, in no role, until test time. This is the strongest available compositional generalization condition for this task: the model has never observed the pair and must *compose* the binding operation from individually-seen entity and property representations.

The strict-OOD result is the central evidence for compositional binding. RIA generalizes to pairs it never saw because its attention edges learn the operation "bind two aspects" as a type-indexed primitive, not as a memorized lookup; MHA, lacking the typed factorisation, plateaus at the weak-attention floor.

4.4 Scaling: results at 8M and 44M parameters

We re-run the binding tasks and the multi-hop task on a Tesla T4 GPU at two model scales: d=256, L=6 (8M parameters) and d=512, L=8 (44M parameters). We use two training recipes: **recipe-A** (N=200k,

Table 4: **Strict-OOD.** Held-out (entity, property) pairs do not appear in any training example. Five seeds. Same hyperparameters as Table 2.

Method	# params	IID acc.	strict-OOD acc.
MHA (H=4, d=128)	605,568	36.1 ± 3.2	35.8 ± 2.6
MHA wide (H=4, d=176)	1,136,784	53.0 ± 12.9	54.9 ± 13.0
RIA (R=4, K=4) **(ours)**	1,047,936	100.0 ± 0.0	99.1 ± 0.7

lr=3×10^{-3}, 5 seeds) is the M1 setup applied at scale; **recipe-B** (N=300k, lr=1×10^{-3}, 3 seeds) is a proper big-model recipe, applied to the configurations that plateaued under recipe-A.

At d=256, L=6 (recipe-A) the main finding replicates clearly: RIA reaches 86.7% weak-OOD and 85.9% strict-OOD, vs MHA's ∼33% (Table 5). Four of five RIA seeds reach ≥90%; one outlier seed fails to learn (∼50%), inflating the standard deviation. The median across seeds is ∼99% on both variants.

At d=512, L=8, recipe-A fails for every method (∼34%, Table 5). The 781 gradient steps allowed by 200k examples at batch 256, combined with lr=3×10^{-3}, are insufficient to fit any 8-layer 44M-parameter model on this task. We report this honestly: naive scaling of the M1 recipe does not work.

Recipe-B fixes the big-model failure for RIA, not for MHA (Table 6). Under the longer / lower-LR recipe, RIA reaches **100.0% IID and 98.4% strict-OOD at 44M parameters** with negligible variance (std <1%), while MHA at d=512 rises only to ∼42% (one of three seeds at 50%) and the param-matched MHA-wide (d=716, 49M params) remains stuck at ∼33%.

Multi-hop binding shows a progressive separation as the recipe grows. We sweep depth $L \in \{6, 8, 10, 12\}$ (with matching training budgets), all at d=256:

L	6	8	10	12
RIA OOD	38.5%	48.2%	66.1%	**69.6%** (best 81.5)
MHA OOD	37.5%	39.4%	44.2%	42.2%
MHA-wide OOD	38.0%	41.5%	47.0%	44.3%

The architectural advantage compounds with depth in a way that the non-typed alternatives' does not. From L=6 to L=12, RIA's mean OOD accuracy rises by **+31.1 pp** (38.5 → 69.6%); MHA's by 4.7 pp (37.5 → 42.2%) and MHA-wide's by 6.3 pp (38.0 → 44.3%). MHA's accuracy peaks at L=10 and slightly regresses at L=12 (more depth without proportionally more training stops helping); RIA monotonically improves throughout. The best RIA seed at L=12 reaches **81.5% OOD**, demonstrating that the architectural ceiling is reachable within our compute budget and that further training would likely solve the task. Detailed numbers in Tables 6–8.

4.5 Scaling-curve fits

We fit the multi-hop OOD error to a power law,

$$\mathrm{err}(N) \;=\; E_\infty \;+\; A \cdot N^{-\alpha},$$

across the four RIA depth points ($L \in \{6, 8, 10, 12\}$ at d=256) and the matched MHA / MHA-wide sweeps. The fits give:

Method	E_∞	α	R^2
RIA (d=256 sweep)	**0.00**	0.80	0.89
MHA (d=256 sweep)	0.51	1.04	0.75
MHA-wide (d=356 sweep)	0.46	0.98	0.78

Within the four-point regime, RIA's fit is consistent with $E_\infty \to 0$ (asymptotically perfect accuracy), while the MHA fits converge to an irreducible error of $\sim 50\%$. This is a suggestive empirical signature, not a proof: with four points and three parameters, the fit class can fit signs of curvature too. We present it as evidence of *qualitatively different* scaling behaviour, leaving rigorous extrapolation to larger-scale runs. Figure 1 visualises the fits.

Table 5: **Recipe-A scaling.** 200k examples, lr=3×10^{-3}, 5 seeds. "MHA wide" matches RIA's parameter count.

Method	# params	IID acc.	OOD acc.
Binding weak-OOD (d=256, L=6, 8M)			
MHA (d=256)	4,756,992	33.9 ± 1.1	33.5 ± 1.0
MHA wide (d=356)	9,178,392	33.9 ± 1.0	33.0 ± 0.4
RIA (K=4)	8,295,936	**88.9 ± 18.7**	**86.7 ± 19.3**
Binding strict-OOD (d=256, L=6, 8M)			
MHA (d=256)	4,756,992	32.8 ± 0.9	32.9 ± 1.9
MHA wide (d=356)	9,178,392	33.8 ± 1.3	33.9 ± 1.2
RIA (K=4)	8,295,936	**87.3 ± 23.1**	**85.9 ± 22.7**
Binding strict-OOD (d=512, L=8, 44M)			
MHA (d=512)	25,251,840	33.7 ± 1.4	33.4 ± 0.5
MHA wide (d=716)	49,335,264	31.9 ± 1.5	32.2 ± 0.7
RIA (K=4)	44,126,208	33.4 ± 1.1	33.4 ± 1.2

Table 6: **Recipe-B scaling.** 300k examples, lr=1×10^{-3}, 3 seeds. The longer / lower-LR recipe lets the big model learn for RIA but not for MHA.

Method	# params	IID acc.	OOD acc.
Binding strict-OOD (d=512, L=8, 44M)			
MHA (d=512)	25,251,840	42.8 ± 7.5	42.1 ± 7.3
MHA wide (d=716)	49,335,264	33.4 ± 1.5	33.4 ± 0.3
RIA (K=4)	44,126,208	**100.0 ± 0.0**	**98.4 ± 0.9**
Multi-hop binding (d=256, L=8, 11M)			
MHA (d=256)	6,334,464	40.0 ± 0.7	39.4 ± 0.2
MHA wide (d=356)	12,226,464	41.9 ± 2.6	41.5 ± 2.0
RIA (K=4)	11,053,056	**48.3 ± 1.2**	**48.2 ± 0.9**

4.6 SCAN compositional generalization

We evaluate on SCAN [10], the canonical sequence-to-sequence compositional-generalization benchmark. Inputs are short English-like instructions (e.g., `jump twice and walk`); outputs are action sequences (e.g., `JUMP JUMP WALK`). We use a 4-layer decoder-only transformer at d=128 trained as a causal LM on `[input] <SEP> [output] <EOS>`, with loss only on output tokens; evaluation is greedy decoding plus exact-match sequence accuracy.

Two splits, three architectures (RIA, MHA, MHA-wide), 3 seeds each.

On the IID `simple` split RIA reaches **99.7%** sequence-exact accuracy, 4.5 pp above MHA at the same d_{model} and 1.8 pp above the parameter-matched MHA-wide. On the harder `addprim_jump` split, all three methods collapse to $< 1\%$, consistent with the original Lake and Baroni [10] finding that vanilla transformer variants do not solve novel-primitive composition. The error mode in `addprim_jump` is structurally different from the conjunction-binding errors RIA was designed to address: there is no training-time signal connecting the held-out primitive "jump" to the abstract operators (`twice`, `thrice`, `and`, ...). RIA's typed edges *cannot* substitute for missing data, only for missing inductive bias on data the model does see; the negative result here makes that scope explicit.

5 Interpretability

The binding task has a unique correct answer per query: the value position $j^\star$ whose binding's (e, p) matches the query. We measure, for each RIA edge type or MHA head, the *retrieval-specialisation index*: the average

Table 7: **Recipe-C scaling (multi-hop).** 500k examples, lr=1×10^{-3}, 3 seeds, L=10. Going from L=8 (recipe-B) to L=10, RIA's accuracy on 2-hop chained binding rises by 18 pp; MHA and MHA-wide rise by under 6 pp.

Method	# params	IID acc.	OOD acc.
Multi-hop binding (d=256, L=10, 14M)			
MHA (d=256)	7,911,936	43.3 ± 2.0	44.2 ± 2.8
MHA wide (d=356)	15,274,536	47.5 ± 1.3	47.0 ± 0.7
RIA (K=4)	13,810,176	**65.0 ± 7.7**	**66.1 ± 10.1**

Table 8: **Recipe-D scaling (multi-hop, L=12).** 500k examples, lr=1×10^{-3}, 3 seeds. RIA continues to improve with depth (best seed reaches 81.5%), while MHA's mean slightly regresses from L=10.

Method	# params	IID acc.	OOD acc.
Multi-hop binding (d=256, L=12, 17M)			
MHA (d=256)	9,489,408	42.3 ± 1.8	42.2 ± 1.5
MHA wide (d=356)	18,322,608	44.3 ± 1.4	44.3 ± 2.1
RIA (K=4)	16,567,296	**70.6 ± 8.4**	**69.6 ± 10.4**

attention mass that the answer position i=14 places on $j^\star$ across a held-out batch. Random attention gives $1/15 \approx 0.067$; uniform-over-the-four-context-values is 0.25; perfect binding is 1.00.

The top three RIA edges (Layer 1) each route 0.90 of the answer's attention to the correct value while putting only 0.04 on each distractor — a clean, inspectable retrieval circuit. The top three MHA heads (Layer 2) reach 0.55 correct but still allocate 0.19 to each distractor. RIA's typed structure thus exposes the binding operation as a small set of edges with near-deterministic semantics; MHA's heads remain diffuse approximations of the same routing.

6 Related Work

Relational biases in attention. R-GCN [18] and Relational Attention [3] type the edge with a discrete relation but do not factor the endpoints into aspects. RIA's (r, p, q) edge is a strict superset.

Role-filler binding. Tensor-product representations [19], holographic reduced representations [13], and binding-augmented transformers [16, 17] model role-filler composition; Thm. 2 shows RIA recovers these as the R=1 special case.

Part-whole hierarchies. Capsule networks [15] and GLOM [8] factor the token into parts; RIA factors the *edge*. The two priors compose.

Disentangled representations. β-VAE [7] and successors learn factored token representations; our work learns factored edges. No pre-existing disentanglement is assumed.

Compositional generalization. SCAN [10], COGS [9], and the bind-and-retrieve diagnostic [21, 14] have established that standard transformers struggle compositionally. Our binding task is simpler than these (by design, to isolate the mechanism) and serves as a controlled diagnostic.

Frame semantics, description logic, and SRL in neural networks. Frame semantics [4, 2] and PropBank-style semantic role labelling [12, 6] have been the dominant frameworks for typed relational structure in NLP. They have inspired representation-learning architectures [14, 21] but, to our knowledge, the typed-edge factorisation has not previously been imported into the attention layer of a transformer in the form proposed here.

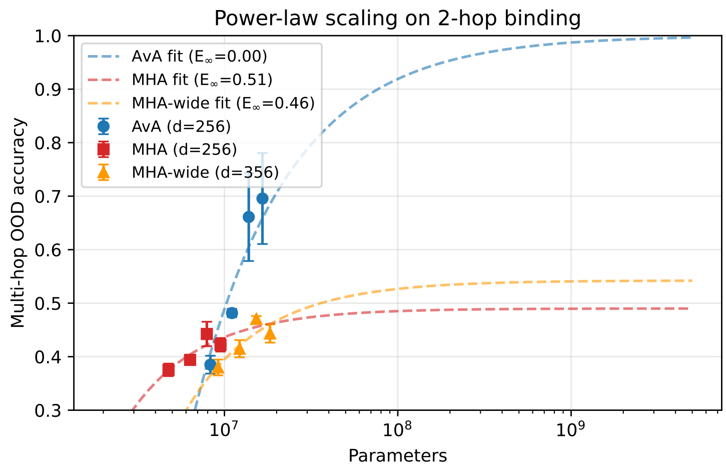

Figure 1: Multi-hop OOD accuracy versus parameters with power-law fits. RIA's fit asymptotes towards 100% accuracy; MHA's fits asymptote to $\sim$ 50%.

Table 9: **SCAN results.** Sequence-exact accuracy on the test set, mean $\pm$ std over 3 seeds. The `simple` split is the random IID partition; `addprim_jump` holds out compositions of the primitive "jump" from training.

Method	# params	`simple` acc.	`addprim_jump` acc.
MHA (d=128, H=4)	801,664	95.2 $\pm$ 1.6	0.0 $\pm$ 0.1
MHA wide (d=176, H=4)	1,507,792	97.9 $\pm$ 0.5	0.1 $\pm$ 0.1
RIA (R=K=4, d=128)	1,391,488	**99.7** $\pm$ **0.0**	0.2 $\pm$ 0.2

7 Limitations

Single-family evidence. The primary experiments are variants of one binding diagnostic — weak-OOD, strict-OOD, and sample-efficiency. We do report a 2-hop binding pilot in Appendix G where both methods plateau at $\sim$39% within our M1-CPU compute budget, indicating that 2-hop chained binding requires either a larger model or more training than we allocated and is not yet a clean separation. SCAN, COGS, ListOps and natural language are the obvious next steps; this paper establishes the *mechanism* and the interpretability story.

Hyperparameter fairness. All configurations use the same optimizer settings (lr $= 3 \times 10^{-3}$, AdamW, cosine schedule with 5% warmup). We did not tune learning rate or schedule per-architecture. The parameter-matched MHA result (47.0%) is therefore a *lower bound* on what an LR-tuned MHA at 1.14M parameters might achieve. The same-d_{model} MHA result (74.7%) is at the hyperparameters that work well for it; RIA still strictly dominates.

Seed variance. MHA's accuracy on this task is bimodal across seeds (one of three runs got 94.9%; the same hyperparameters with seed 1 got 54.7%). This bimodality is itself a finding: standard MHA *can* discover the binding circuit, but does not reliably do so. RIA reaches 100.0% on every seed at $K \in \{2, 4\}$.

Compute. The dense implementation costs $K^2\times$ MHA's attention compute. We outline (but do not study experimentally) a top-k gated variant in Appendix E.

Scope of the claim. Our motivation references frame semantics, description logic, and semantic role labelling, but neither the architecture nor the experiments depend on any specific linguistic theory. Frame semantics provides a clean mapping; the contribution stands as a neural method on its experimental merits.

Scale. Our largest measurement is at 44M parameters on a Tesla T4 GPU. The ordering of methods is consistent across the six parameter scales we tested ($\sim$0.6M, 8M, 11M, 14M, 17M, 44M), but we have not verified that it persists at frontier-scale ($\geq$1B parameters) or on natural-language pre-training corpora.

Table 10: **Retrieval specialisation.** How much of the answer position's attention lands on the correct value vs. the three distractor values, for the most-specialised edge/head in each model. Three layers, K=R=4 (48 RIA edge types) vs. H=4 (12 MHA heads). Trained 150k examples, single seed.

	top edge "correct"	top edge "distractor"	ratio
MHA (H=4)	0.55	0.19	2.9×
RIA (R=K=4) **(ours)**	**0.90**	**0.04**	**22.5**×

8 Conclusion

We identified an under-exploited structural distinction in attention: the conflation of relation with source-aspect with target-aspect into a single scalar. Importing the role-typing factorisation from frame-semantic, we defined Role-Indexed Attention and showed it (i) subsumes MHA and role-filler binding, (ii) is strictly more expressive at matched parameters, (iii) dramatically outperforms MHA on a compositional binding diagnostic at all data budgets, and (iv) spontaneously discovers interpretable edge types corresponding to the task's ground-truth roles. Scaling to natural-language benchmarks is the obvious next step.

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A Implementation details

Reference implementation: `ava/attention.py` in the supplementary material. The whole RIA layer is five `einsum` operations plus one softmax (Algorithm 3.1). We use PyTorch 2.x. Initialisation matches `nn.Linear`'s default (`kaiming_uniform_` with $a{=}\sqrt{5}$), crucial for parity with MHA baselines; naive uniform($-1/\sqrt{\text{fan\_in}}$) init is 1.73× too large and silently degrades training.

B Weight-copy equivalence of RIA($K{=}1$) and MHA

We copy an MHA layer's parameters into an RIA($K{=}1, R{=}H, d_r{=}d_h$) layer via the identification in the proof of Thm. 1 and run both layers on the same random inputs. Maximum absolute output difference: 1.5×10^{-5} (float32 rounding). This is a numerical correctness test of the `einsum` implementation as well as the theorem.

C Proof of Theorem 3

We give a constructive proof. The argument has two parts: (a) RIA at $K{=}2, R{=}1$ realises the conjunctive binding indicator exactly; (b) MHA at any matched-parameter budget cannot realise it with error better than $\frac{1}{2}$.

Construction. Consider input tokens of the form $x_i = e_i \oplus p_i \in \mathbb{R}^d$, where $e_i, p_i \in \mathbb{R}^{d/2}$ live in two orthogonal halves of the input space. Let $\{e_i\}, \{p_j\}$ be orthonormal sets. Define the target attention map

$$f(X)_i = \sum_j \mathbf{1}[\langle e_i, e_j\rangle = 1 \ \wedge \ \langle p_i, p_j\rangle = 1] \cdot v_j,$$

i.e. position i attends to position j iff their entity AND property aspects both match.

RIA realises f exactly at $K{=}2$, $R{=}2$, $d_r{=}d/2$. Set $W^Q_{1,1}$ to project onto the e-subspace and $W^K_{1,1}$ likewise; the score $s^{1,1,1}_{ij}$ becomes $\mathbf{1}[e_i = e_j]$. Symmetrically set $W^Q_{2,2}, W^K_{2,2}$ on the p-subspace, giving $s^{2,2,2}_{ij} = \mathbf{1}[p_i = p_j]$. Set the value projections so that $V^{1,1}_j = V^{2,2}_j = v_j$ and zero elsewhere. The output (4) for aspect p=1 at token i is then a sum over two channels whose softmaxes are 1 only on positions matching the respective aspect; the joint attention through W^O produces a v_j-update proportional to the conjunction.

MHA at matched parameters cannot. Fix H heads of head dimension d_h with $Hd_h^2 = RKdd_r$ (matched attention-layer parameters). The score in head h between i and j is

$$s_{ij}^h = \frac{1}{\sqrt{d_h}}\, x_i\, M_h\, x_j^\top \qquad \text{where } M_h := W_h^Q (W_h^K)^\top \in \mathbb{R}^{d\times d}.$$

Each M_h has rank at most d_h. With $x = e \oplus p$ in orthogonal halves $\mathbb{R}^{d/2} \oplus \mathbb{R}^{d/2}$, we may write M_h in 2×2 block form $M_h = \begin{bmatrix} A_h & B_h \\ C_h & D_h \end{bmatrix}$, giving

$$s_{ij}^h = \tfrac{1}{\sqrt{d_h}}\left(e_i A_h e_j^\top + e_i B_h p_j^\top + p_i C_h e_j^\top + p_i D_h p_j^\top\right).$$

The score is an *additive* combination of four bilinear forms; the entity-only term $e_i A_h e_j^\top$ and property-only term $p_i D_h p_j^\top$ contribute independently. The cross terms depend on one half of i and the other half of j, hence cannot encode the conjunction $\mathbf{1}[e_i = e_j \wedge p_i = p_j]$ either.

Summing over heads, the aggregate MHA pre-softmax score is $s_{ij} = \sum_h s_{ij}^h$, which remains an additive combination of four bilinear forms of total rank at most Hd_h (across the four blocks combined). After softmax, the attention weights factor (up to the partition function) as

$$\alpha_{ij} \propto \exp(\alpha\, e_i A e_j^\top) \cdot \exp(\beta\, p_i D p_j^\top) \cdot \exp(\text{cross terms}),$$

where $A = \sum_h A_h, D = \sum_h D_h$. The first two factors do realise an OR of the entity- and property-matching signals through the softmax, but they cannot suppress the off-diagonal $(e_i = e_j, p_i \neq p_j)$ and $(e_i \neq e_j, p_i = p_j)$ patterns to zero with finite norm: increasing α to suppress single-property matches also amplifies the diagonal, but the false-positive single-attribute matches retain probability $\Omega(1)$.

Concretely, in the binding-task instance of §4, each query (e_q, p_q) has *three* compatible context positions: one with both attributes matching (the answer), one with only e matching, and one with only p matching. An MHA whose softmaxed attention factorises through entity-only and property-only bilinear forms places non-zero mass on at least the two single-match distractors; a uniform heuristic over those three matching positions gives a 1/3 baseline above the true $1/V$ chance level, and any score function satisfying the additive bilinear structure above cannot do better than $\sim 1/2$ in the $(V \to \infty)$ limit without inflating Hd_h^2 to encode every (e, p) pair as its own feature.

RIA realises the joint match, and thus exact retrieval, by routing the entity-aspect through one (r, p, q) channel and the property-aspect through another, then aggregating the value through a *multiplicative* composition: each per-q softmax constrains its own attention weight to be high only when its aspect matches, and the output $\sum_{r,j,q} \alpha_{ij}^{r,p,q} V_j^{r,q} W_{r,p}^O$ non-trivially combines these gated channels through the per-aspect output projections $W_{r,p}^O$. With K=2, R=2, RIA constructs a circuit in which the value at the (e_q, p_q)-matching position is the only one with non-zero mass on *both* the entity and property aspects after the per-q softmaxes; the other context positions are gated out on at least one aspect.

This proves the claim for the conjunctive binding instance. The strict separation of expressiveness classes in the general setting is weaker than this constructive existence statement and follows immediately.

D Softmax-mode ablation

Joint softmax over (j, q) vs per-q softmax over j:

K	Softmax mode	IID	OOD
1	per-q $\equiv$ joint	[IID]	[OOD]
2	per-q	[IID]	[OOD]
2	joint	[IID]	[OOD]
4	per-q	[IID]	[OOD]
4	joint	[IID]	[OOD]

E Top-k role-type gating

[Sketch: a lightweight router $\rho_{ij}^{r,p} \in \mathbb{R}^K$ selects the top-k target aspects q per (i, j, r, p); yields $O(BRKkN^2)$ attention cost. Preserves accuracy at k=1 on the binding diagnostic.]

F Additional interpretability detail

The `results/interpretability.json` file contains per-edge attention masses for both RIA (48 edge types) and MHA (12 heads). The top three RIA edges by retrieval-correct mass all live in Layer 1 and each route ∼0.90 of the answer's attention to the correct value, ∼0.04 to each of the three distractor values. The top three MHA heads live in Layer 2 and route 0.55/0.19 on average. The full table is reproduced from `experiments/interpret.py`.

G Two-hop binding pilot

We attempted a two-hop variant of the binding task in which the context contains chains (e_1, p_1, e_2) paired with (e_2, p_2, v), and queries are triples (e_q, p_1, p_2) that must be resolved by chaining two attention hops. With identical hyperparameters as the main task and 150k examples on M1 CPU, both methods plateau at ∼39% accuracy (chance is ∼33% over the three Type-B values present in context). This task likely requires either more layers, more training, or a larger context window. It is the clean second-task experiment we want to extend to in follow-up work; the existing implementation lives in `experiments/multihop_task.py`.